

ActInf Textbook Group ~ Cohort 1 ~ Meeting 17

(Chapter 7, part 1)

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welcome it's cohort one meeting 17 already and it's September 23rd 2022 we're having our first discussion of chapter seven in the textbook well we have many ways to go we have questions on chapter seven that people have been adding we also have um some summaries and overviews and uh can walk through the text directly but first does anyone want to just raise their hand or unmute and give any thought that they had on chapter seven their experience reading it their understanding of where it's situated in the textbook overall what was in it or what was not in it Etc laughs okay well all uh open again please raise your hands or write in the chat if you want to address anything at any point I want to open um by acknowledging a lot of the contributions ali uh made and uh in some conversations that he and I had earlier this week that I believe all of you cohort wonders will find interesting um but first let's just start with this quotation we're going to start with the opening quotation the chapter then I'm going to surface some discussions with Ali and then we're going to go into some details of the chapter so the quote is what I cannot create I do not understand by RF so what does anyone think about that quotation or what is it uh mean in this context um I think uh I mean it it's more applicable applicable applicable the more complex the the thing is that you're trying to build but like how could your generative what priors or for what way could your generative model create and accurate prediction of the operation of something or the underlying dynamics of something if you do not kind of step-by-step generate the afford and take the actions that generate those affordances I don't see any causal path to doing so with help like you would have to step through it necessarily if it's computationally like complex so that's most things in the world but yeah awesome it's like this is like the low road answer like how can the generative model have anything like understanding without generating you can't just have it on the Shelf and then someone um has added the mild answer we can't have understanding just through mental envisioning that the algorithms need to be implemented for a learner's Journey and then this is also um an even deeper or stronger point which first and

others have been working on for a long time which is like it's sentient artifacts in the world that will be the realization of active inference it's not just like some nice derivations Brock and then anyone else I was just gonna add uh something about hidden states there I mean it's again coming back to computational complexity but there's there's things now that we're starting to build that we don't really 100 understand all of the Dynamics of what is or just in general like we invent things that we don't completely understand first and then based on the observational based on the evidence that we observe that it is consistently exhibiting some behavior that that directs our attention that directs where our generative model pursues like you know more observations but um we're definitely getting towards a point where that's like kind of no longer going to be possible in the way that we usually use them try to use math to just shortcut stuff where the the operation of the Dynamics of the things that we're going to build are necessarily going to be they will be the proof of their existence will be their own proof awesome okay so just fun you know starting quotation upload that one for sure though um there's some detailed things about [Music] the examples and I also uh and some of these I've been working on everyone is welcome for every chapter to be contributing on these Pages like kind of just trying to overview what these examples are because a chapter it states it up front yet I missed it the first several times reading it like these Illustrated models of every color in the rainbow are the section titles and those are the functionalities that are getting layered in sometimes building on the same model like there's like teammates version one then there's team A's version two other times it's switching between models or showing two different models that illustrate the same cognitive functionality other times the simulation outputs are shown other times they're not so there's like a lot of heterogeneity in the observations in this chapter but it really wasn't until I saw that these first paragraph words were the section titles that it made more sense how the chapter was being like laid out um so that's just kind of one uh notes Ollie do you wanna raise any thoughts or describe a little bit about our discussion on like how we're going to move forward on the equations okay if you unmute then go for it otherwise I'll um let me see okay yes no Mike thank you all good so um we were discussing how there are several layers um as with other scriptural Traditions there's levels of readings so there's a level of reading that is even um more granular or below where we've been currently targeting like here's equation two five there's a reading for accessibility and understanding that would potentially be like $f \text{ open square bracket } Q \text{ comma } y \text{ closed bracket equals negative sign expectation over } Q \text{ open parenthesis } X \text{ Etc}$ so it's like the reading of the notation then the notation can be substituted for terms in the active fontology which is like the level that's being

described here but this is not to be seen here this is the level where notation could be aligned across papers because their notational reading would be different but then they might be referring to the same exact composition of the active ontology then once the ontology definition has been squared the terms could be condensed into meaningful units like accuracy complexity risk expected risk that gives us a symbolic in the more conversational sense rather than in the like notational sense and then that's where memes and themes and rhetoric around like what active inference formalisms say start to arise and then there's the consequences of changes in relationships within and outside the formalism because it's kind of like we're all there's this larger structure for like a pomdp and changing one variable if you're only looking at like one view or one kind of sub formalism but you know turn it up to 11. yet that might influence other variables that aren't part of that formalism itself and so this is like a very open question that touches on everything from like the implications of changes to relationships that aren't formally specified for example the base graph the edges reflect a certain type of relationship as we've explored but changes from one can propagate throughout a whole system potentially in non-linear ways so it's always going to be this kind of open area how changes including like counterfactuals about the formalism what if risks were situated this way instead of that way or how does that relate in this case like so this is kind of an open one again people please update it and and modify it um but like some of these equations which we'll get to so we'll come to why we had that conversation in a 7.8 later but that was like one um really conversation and then there was one other then just to kind of jump into a chapter seven topic but also kind of reframe our Learning Journey as we're all working through this for the first time um in the textbook group so they wrote this choice in the team Ace and then Eric or anyone else I know that you had some questions on the teammates um or we can go to the earlier examples this choice of what to do do you seek out the informative cue and then go to the arm that you've now reduced your uncertainty about where the reward is so do you take an epistemic action and then have like a better chance of making the pragmatically good choice or do you just you know go and make the pragmatic Choice do you take one step up or do you take one step back for two steps up it speaks to the exploration exploitation dilemma in Psychology and machine learning a dilemma that's resolved under active inference so we were just cracking up because first a low familiarity learner might not know what these terms are or like why it's relevant to even talk about this situation familiarity might look like knowing metaphors examples intuition some relevant citations or historical anecdotes about this the understanding in terms of what the active inference formalism says about this you know if it's

resolved under active inference then understanding that resolution is about the basis and then from understanding there could be like further developments and applications um and then we'll we'll come to this but what was making us laugh was surely there are parameterizations in which the rat and the teammates always does one thing or always does the other so it's not that the explore exploit trade-off is simply resolved under active inference even though people commonly point to things like the ability to rapidly switch between exploratory and exploitative behavioral policies because there's parameterization of that model and structure learning including that that has to be accomplished such that the model can be in a kind of critical point where it behaves adaptively and thus in a situation manages the explore exploit tension under certain constraints but how could it be said that active inference resolves that dilemma so maybe to um whomever wrote this one what is happening yes please Eric click classical Eric writing here so what what about this uh yeah I'm I'm kind of the Heckler at the garden party here you can see in most of my my comments and questions but yeah I mean maybe it's just best to I mean it's maybe a bit of red than spoken but yeah I don't I don't I don't see this as a valid claim that they well first of all it's not a I don't know why they call it a dilemma for psychology it's a trade-off it's been understood for years and there's ways of formalizing it and you solve it through optimization it might be a dilemma for a creature because they don't they don't know you know what they should do but then but that's not a dilemma for the field and then uh I guess the other other paragraph is I I don't see that they offer any resolution all they do is they put it into the same they use a Common Language free energy they put exploration in and exploitation into the same equation and call it you know with this information based term called free energy which is perfectly valid that's great but that doesn't resolve the problem because as you pointed out you've got parameters and uh the designer or somebody has to figure out what those relative weights are the parameters are and uh so you really have an advanced any you know fundamentally simply by using this free energy way of expressing what the trade-off is as opposed to some arbitrary measure called energy or preference or something like that so I just think that they're that it's it's it doesn't help them when they over claim which is my opinion man you could convince me otherwise I'm open um but it seems to me it doesn't help them by overstating their contribution to resolving exploration versus exploitation yes very insightful I actually totally agree I purposely tried to not mention it as a dilemma either I think tensions trade-offs um these are all valid um but but that's that's uh like kind of micro linguistic and a more serious issue is that claims that do crystallize out from the text and things that people can copy and paste and quote

you know if this were a um a courtroom scenario I mean did you say that active inference resolves exploration exploitation it's gonna have the answer has to be yes it was claimed in the book could somebody contextualize that like it's providing a first principle's way where there are tunable or learnable or parameter sweepable um features of models that sit at a really well positioned uh intersection of human interpretability and manifolds of relevant model variation yes in an extremely charitable reading one could kind of bring this up from back from coming off over the edge back onto the table that would also um make active inference less surprising and less realistically hyperbolic and be part of this process of seeing the evidence in the realization not in the um kind of ironic like we have a different partitioning of a statistical variable value so nice points um Brock um I guess I wanted to agree with that and then also maybe play a little Devil's Advocate because it's like uh yeah we're all it's all information it's solved um that's not really just because it's been formulated or renamed to something pragmatic value epistemic value it doesn't seem like that resolves it or something I'm wondering if the reading of that though could be that not that there is a TR there is still a trade-off but that the Dilemma is that what they're talking about is that the trade-off like when do you explore or when do you exploit not that there is a trade-off but that that is uh this last sentence and a half here of um or the resolution stems from the minimization of free energy is that whether you would seek pragmatic value or epistemic value is conditioned on which you your model believes to be the free energy minimizing choice though it doesn't you still have the Dilemma so to speak um the trade-off but which side of the trade-off you take is maybe result by generative model here or not I don't know is that a day sprock I think the field is set for a situational resolution that may be as good or maybe better than other ways that it's been addressed it it comes down to the exact numbers that are chosen so like in the example of the teammates um so we can look at for example this 0.6 and negative 6. huh what if it was negative 3 000 and positive three thousand and so this came up in the um live stream 45 with Ryan Smith on the folk psychology where the intensity of the preference was represented and so if the preference is so if you're okay we prefer having more food is that going to be a thousand versus zero because if so even the scantist probability of achieving food will be pursued in other words that model is going to be parameterized way way way on one side and it's going to exploit only or it's only going to take locally um greedy Behavior whereas if someone said well my behavior for food is point zero zero zero zero one then then maybe even the most obvious strategy would not be undertaken because it's so close to having a flat preference Y is 0.01 versus 3000 why is three thousand too high and 0.01 too low well it has

to do with like the ratios and the interactions of a lot of the parameters so it's it's not even going to be like well my preference is three comma one that is going to be just like two parameters being drawn out from a potentially massive parameterization of a model and so preferences of just only speaking of preferences maybe um very hard to compare across situations because it's not like plus six for the preference for food and negative is it doesn't have a semantic meaning in the world it's literally a model parameterization then the question becomes how and this framework or this way of um having of combining probabilistic inference and energy-based inference for strategic decision making how can those parameters be tuned into a region of Subspace or manifold where the behavior is flexible and adaptive so I think the stage is prepared for a first principle's resolution in very charitable reading sorry yes please yeah I was just going to say Ali shared this paper meta control uh of the uh exploration exploitation it's in the nearby okay chat link um nice but that yeah that's that is it's it's not that it solves it like in some Grand it eliminates exploration exploitation or that it even completely currently yeah it's it's just the affordance I guess seems to be there to explain which one why like you're going through that you know there's a preference heavy here or heavy there and it's you know not that it's it's optimal right it's not not that it makes sense or it's a depth adaptable or anything like that necessarily just that there is some causal through line Maybe yes Eric yeah um so I want to first read Ali's comment here um the critical point to establish is to Define um action selection as a balance between maximization of expected reward and expected Information Gain which are functionals of posterior beliefs about uh late states of the world um so that touches on a different one of the questions which I also wrote comments on which is about the team A's example in particular and we'll we may I don't know if we'll have time but we may get to that one today so but the questions specifically I I want to raise about this comment or maybe this I guess is a quote about there being a balance between maximization of expected reward versus expected Information Gain that's one way of having a trade-off but you might look for a a um an agent that only wants to maximize reward in the long run but in order to do that that entails Information Gain and that therefore gives a motivation for why you want to um attend to the other term the the the information based you know information or learning based term or knowledge uh refining of State base based term but that's in the service of the larger gain and that's essentially what a um uh reinforcement learning is tries to do is say look we're going to look ahead so that uh the moves we take now will bring us advantage in the future and it's you know there's a time discounting and all that so that doesn't that that takes away the idea of information it's as being a value in and of itself

but it's only for a purpose and so I you know well there's a specific you know libraries that specifically with regard to the team teammates example and a different question um thank you the balance between pragmatic and epistemic is the optimization so that is the question and that is the critical point and that's the situational balance that bottling is going to be evolving around it's like some situationally um optimal or preferable strategy around that is the point but pointing to the point is not the does active inference provide us a natural language grounded first principles approach it may that is also not dilemma resolution so on to some questions but these are really great points um this is uh just an example that quite of a type of question that I encourage people especially if they're going back to the earlier chapters or just whenever they feel like making a contribution of this kind these are like super helpful questions again the questions we write some of them may be included in really important educational materials not the least of which many further textbook groups but potentially even far beyond so what would you ask somebody to understand their comprehension of the materials so I just said what are the rows columns and numbers in these equations and it even says it in the text it's the probability of the next state so the next state is in the rows and then the current state is in the columns and this gets distributed across everything so that the rows and columns sum to one as any other transition probability Matrix would be expected to have um so we don't need to discuss it um unless somebody wants to go into a little more detail but this is going to be at the core of that third step of the recipe which is you might have the structural form from the second step of the recipe in chapter six and then the next step is to kind of specify or instantiate like in code what the dimensions are of these matrices this is just saying like they are matrices there's some prior tensor Matrix some hidden State some transition Matrix some emission Matrix and this is like the structure step two of the recipe step three how many rows and columns and as Jacob and others have been exploring there's the rows and Columns of the analytical representation with the and then there's how Pi mdp does it and there's how different computational realizations do it um Jacob would you want to add anything about like the dimensionality of these matrices it could be interesting yeah I mean uh I can probably speak more to how Prime DB deals with the dimensionality which currently is probably um unnecessarily as unnecessary like redundancy to them uh like if you're dealing with more complex State space or rather more High dimensional State space and you have different modalities that you want to encode in these matrices you uh the because you're then uh performing Matrix multiplications you need to keep the dimensions of the matrices in those given modalities equal at least in time DB not in not in methods as far as I'm aware um but

that means that then you have Dimensions Within These matrices that don't contain any information so say you have a a grid that you represent for the likelihood mapping of observation and uh state state in physical world but then you have another um another type of observation like where is another agent located with respect to you like um are they in a certain direction and that might be represented by uh five observations say like uh they're above they're below they're to the right to the left or not at all then you would have then you would need to encode the observation in the 9x9 Matrix but then you need to replicate it five times because it's that is the same for each of those relative observations but then when you want to encode the likelihood mapping just for the second type of observation uh I'm actually not entirely sure what that would be what I'm presuming it's going to be like five by five but then replicated nine times or something like that just so the dimensionality matches because then you're doing inference over both modalities and at least in Prime DP you're then performing Matrix multiplications over both of them thank you Jakob one way I'm kind of like seeing that is there's the sparsity of the graph but then in time DP or in in any given computational implementation the data structures including potentially like auxiliary data constructs may have a dimensionality that reflects potentially at worst a combination of the dimensionality of other aspects as part of even just a temp file that you didn't even exactly specify there might be some intermediates stated or unstated intermediates whose dimensionality isn't merely the dimensionality of the analytical representation it's a little bit of a subtle implementational point but this is the mountain to climb to implement these kinds of models and also just while we're on this topic I think it's one area where CAD CAD is going to be very interesting to explore in octoblock fronts package because we also can imagine that the execution order of certain operations even in a single agent setting not just the multi-agent setting is really relevant so there's that question all right so figure 7.2 all right so the first example is about a musician playing music we are not aware of this example being used anywhere else like the citations following those this musical note type representation is not like shown um and previously we kind of discussed how some of these Trace black and white images are really um ambiguous because of being like intersecting black lines and then of course the um the empty Square so yes can you say where those lifelines come from you want to dive into that right now um well the upper left each block line even though it can be hard to trace them I think they're just being shown like representationally their beliefs about each note in the sequence at each time step so the time steps are on the x-axis and the um the uh belief in terms of a probability are on the y-axis so initially right so there's some there's some sort of

and this gets to another question that was raised but maybe this is time to talk about it there's some sort of engine some sort of differential equation same you know simulator or something running underneath here right that is not discussed in the chapter but that's generating these belief curves yes yes it's related to some functions in Matlab and I was just going to mention that because my sense from the discrete model would be you'd have five let's say points on time zero then you'd have five points on time one so I thought well what is happening with this kind of like you know little fractal crash right here I mean what how could you have a continuously differentiable um oscillating type Behavior when all you're calculating analytically is just one two three four five then I thought is it meaning is is the are the points actually at the midpoint and there's a differential equation that's sort of like ribboning an action perception Loop through a discrete time Matrix but then if this is a function that's evaluable at every point aren't we in a continuous time setting so wouldn't that have to imply like a machinery for unpacking a continuous time action perception cycle from discrete time specification so I agree very much yeah good yeah my guess is what they do is they they've got their um you know these equations if the free energy equations and the the parameters are these distributions okay what's your belief in these different properties and um those parameters evolve over time through a differential equation because you turn this you turn these things into a differential equation and when you change when you change the state something like okay now we've got another note comes in then that's what's going to trigger um the differential equation to go in a different direction essentially and that's what that's why you've got the five discrete regimes here or you know um Target Target points that it kind of evolves to um and then you know the next note comes in and it now the boundary conditions change essentially so your differential equation takes you in a different direction so I and I I recall that there's something like um you know you guys probably discussed this more more than I paid attention but there's some sort of universal solver that they're providing um so you just plug in your um State your your Matrix matrices your State equations or your transition matrices and then it'll solve it all for you and that may be what this differential equation solver is about you think that that sounds right yes I think the um yes when whenever variously in the method sections and supplemental sections of paper they'll say here in the paper we're only specifying the generative model and the matrices and then we use standard routines to address it now I don't think that has the um it's not the Pinnacle of accessibility and reproducibility however it does at least use standard routines but again those are not always specified in the paper which is um an issue so it'll be important to regenerate examples within SPM and then

also to use some of these things that we were discussing with like the ability to move between different languages um but like vbx variational Bayes X I don't know what the x is actually for um it is very it's one of the core functions it is actually doing the variational inference in SPM and again SPM is like this sort of like Chimera package because it arose from the immediacies of needing to do neuroimaging registration and dynamical Analysis that then broached into Dynamic causal modeling um random field Theory based statistical testing permutation testing all these areas that are not um themselves formally linked per se but rather useful as part of the toolkit of a neuroimaging researcher then just as we were a little bit discussing in um like chapter two and of course many times earlier in that generalized Bayesian inferential framework hierarchical modeling including action is actually not the hardest thing yes implementing action is one thing but treating action selection as inference planning as inference it has some uh challenges that arise relative to Just Doing Time series anticipation um but then those functions became compiled into SPM in a limited capacity um so they again they go through some of this but but I think it's it's a key issue um Ali we'll definitely look forward to what the um free energy gradients are but like what where the continuous nature of these curves and like why there are the numbers that there are one two three four five six seven eight nine ten now there's five notes ten is two times five like it's just there's a little bit more that needs to be fleshed out there and then one question that was like again somewhere between testing or comprehension and being clear about what we're doing here um and in this ambiguity we can ask it without any bias um the negative free energy gradients I.E prediction errors uh what is a prediction error a free energy calculation well ignore a gradient for a second free energy calculations are not prediction errors prediction errors are observations minus expectations so this is like some finite value in the state space of what is being measured free energy is not that it references potentially those variables but it does things that we know about like KL divergences and such okay so free energy is not prediction error and that's a huge difference between for example a um predictive processing framework and a free energy hierarchical predictive architecture Eric but they're saying the gradient and free energy is a prediction error exactly so even if it were that's then that was that was the next question which was what's the difference between free energy and the free energy gradients are we talking about a landscape of derivatives of free Energies or are we talking about what here like is it being described as a gradient as a and hence gradient-ish because it can be calculated over some discrete bi-continuous State space it's like five skyscrapers that you could have derivatives over in the y direction but you couldn't take the partial X or is it truly um or is it a

continuous landscape that's just being tethered to the discrete values in X but then again that's the whole continuous discrete question or are the values gradients which has an even more complex interpretation we don't have the answers on these but the these um it's hard to understand how these figures could have epistemic or pragmatic value for Learners or practitioners without some of these questions being resolved all right um the first example this is just kind of from this from walking through so the first example was just hidden Markov model hidden States updating Through Time Bayesian filter common filter this is like just standard signal processing and a Bayesian framework um and we're going to see a lot of Echoes of the step-by-step model stream one they give an even simpler example with static perception One Step perception um go second to the dynamic case then introduce action π policy and then Nuance policy through like uncertainty and so on um so that's the first inferential case they're then going to head into decision making and planning as inference we see policy being introduced as a variable that influences how States change Through Time does policy do that or does Action do in the one step limit policies our actions are affordances but π is specifically reserved for sequences of actions so is it that sequences of actions like do that or is it um you know is this the actual base topology um but nonetheless that architecture allows the evaluation of alternate policies in terms of their relative expected uh variational or expected free energy depending on whether a one-step policy is accomplishable with variational free energy that's like the instantaneous move most consistent instantaneous maneuver versus even one step in advance is going to Broach into this whole expected free energy space um we had on 130 um so uh there's a really uh important discussion um Ozaki is also very interesting researcher neural rhythms and so on factorization is a really important topic and being able to distinguish like neurologically what do we mean when we talk about the what and the where stream the dorsal and the ventral stream and critiques of it Etc but just what and where in the brain and um then in the factor graphs and computationally because the unfactorized models if we're just doing the all by all by all by all those become intractable very fast and the specification of the sparsity of the model is one of the same in the variational base framework as the factorization of the model so factorization is like kind of focusing your search efforts on manifolds where you've constrained certain things to be like linked or unlinked that is what motivates the structure learning problem and the um need to have um to not be locked into factorization schemes at a given level of analysis another just uh question to explore um like here we have the top of a half a figure four three there's just one a matrix so why are there two a matrices here well it's not not an impossible question and the symbols go

a long way here's the four locations that the animal can be in starting position bottom position left arm or right arm here are the the it's five on The X here because that bottom can could reveal an r or reveal an l in this case it reveals an R and the food is honestly there and then this is a second a matrix that is um describing how location is associated with the food non-abversive or positive but this ties into the earlier discussion like it might be all one thing to say well it's um four by five and then four by three but is this actually a four by five by it can't be I I know it's not Eric what do you think yeah um and they mentioned and then there's a tensor also um which and the tensor means that you're stacking this one uh this image against the version where the um reward is on the the left side so the tensor is going to be four by uh eight by two so that makes sense for being the tensor so this the state is another layer on top of this the hidden State there's another layer on top of that section seven three so this would be a fun you know um kind of like PhD qualifying level just describe the whole team Ace every row every column and every value why is it that way so we're not going to type it out here that is to understand that that example and what what the B matrices are and just what what and and to have agility and just identifying the differences between these matrices numerically like pattern recognition like oh here there's two ones here two zeros here two zero zero two ones here one and zero these stay the same and then to be able to transpose that in your understanding to what this means in terms of transition frequencies every place where there's a difference it's a difference that makes a difference so being able to understand what those are is about understanding this example and one can imagine especially if they're wanting to make a application of active inference that's more complex than this 4x4 being fluent with how these matrices are constructed their dimensionality how differences in the generative process are Incorporated how differences in the generative model are Incorporated all these different features are important um this is yes Eric this is definitely very important um question agreed like it it looks extremely neural this has and they these traces come up all the time um similarly here we have three discrete time points starting going to the queue getting the L and then going to the left arm but then what is a step one and a half is the one all of these points and then by two their uncertainty is resolved the belief about certain things goes to zero the belief about the things goes to one and this is just a pure interpolation that is quite a specific interpolation including what appear to be some like dopaminergic spikes or something that are not reflected at either of the time points my guess is that you've got three different blocks there so if you take the one as being the width you know one-third of the way across the two is another third and the three is another third then that switch there that you see happens as soon

as you transition from the one to the two oh although still there's even the graphical interpolation question like and it's just it's so it's it's so easy to be like reading you know just is it just a quirk of the dashed line but is it are there two dashed lines and one two dashed lines diverged in a neural Trace um so it but it's um it's it's kind of clear these ones the discrete formulations are clearer are rats prone to useless Behavior what do you think what useless Behavior I I wrote it out okay okay yeah this is good maybe we'll have time for today but that's my most provocative comment on this one and maybe we can save that for next week okay perfect let's pick up with this one um and any other questions then we'll um glance over the visual cicade look at the learning example consider the hierarchical example there's the maze then there's the hierarchical example another nested discrete meets neural continuous Trace and then we end on 7 squared so thank you all have an excellent day